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ASPS: Augmented Segment Anything Model for Polyp Segmentation

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Abstract. Polyp segmentation plays a pivotal role in colorectal cancer diagnosis. Recently, the emergence of the Segment Anything Model (SAM) has introduced unprecedented potential for polyp segmentation, leveraging its powerful pre-training capability on large-scale datasets. However, due to the domain gap between natural and endoscopy images, SAM encounters two limitations in achieving effective performance in polyp segmentation. Firstly, its Transformer-based structure prioritizes global and low-frequency information, potentially overlooking local details, and introducing bias into the learned features. Secondly, when applied to endoscopy images, its poor out-of-distribution (OOD) performance results in substandard predictions and biased confidence output. To tackle these challenges, we introduce a novel approach named **A**ugmented **S**AM for **P**olyp **S**egmentation (**ASPS**), equipped with two modules: Cross-branch Feature Augmentation (CFA) and Uncertainty-guided Prediction Regularization (UPR). CFA integrates a trainable CNN encoder branch with a frozen ViT encoder, enabling the integration of domain-specific knowledge while enhancing local features and high-frequency details. Moreover, UPR ingeniously leverages SAM's IoU score to mitigate uncertainty during the training procedure, thereby improving OOD performance and domain generalization. Extensive experimental results demonstrate the effectiveness and utility of the proposed method in improving SAM's performance in polyp segmentation. Our code is available at https://github.com/HuiqianLi/ASPS.

Keywords: Polyp Segmentation · Segment Anything Model · Domain Adaptation.

1 Introduction

Automated polyp segmentation stands as a pivotal tool in the diagnosis of colorectal cancer, to aid effective interventions and timely treatment strategies.

Studies like Polyp-PVT[5], SSFormer[22] used Pyramid Vision Transformer for polyp segmentation; CFANet[35] integrated boundaries with a Cross-level Feature Aggregation Network; Endo-FM[23] captured spatial-temporal dependencies to build a foundation model. However, limited by the model's size, the existing methods still lack sufficient capabilities for feature representation and extraction, making it challenging to fully capture the morphology and characteristics of polyps. Furthermore, the limited scale of the dataset may limit the diversity and generalization of the existed methods. Recently, the Segment Anything Model[12] (SAM) was introduced. SAM is pre-trained on the largest segmentation dataset SA-1B, demonstrating remarkable segmentation capabilities across various downstream tasks. With its significant model size and data size, this innovative approach has introduced novel perspectives to the field of polyp segmentation. It also possesses enhanced representation and feature extraction capabilities, surpassing existing methods.

However, SAM's performance in the polyp segmentation task is unsatisfactory [34], due to the domain gap between the training data and endoscopy images. This results in two primary issues: firstly, SAM fails to adequately capture the distinctive features of polyp images, leading to a bias in its learned representations. Secondly, it produces erroneous predictions with inaccurate confidence estimates for out-of-distribution (OOD) data. In addition, because it relied on prompts, SAM has significantly impeded its convenience in clinical applications. Despite several methods improving SAM, such as MedSAM[15], these approaches either rely on prompts or directly fine-tune substantial models. SAMUS[14] effectively integrates CNN and ViT, but its design is quite complex and is particularly suited for processing small images. Consequently, the efficacy of these methods is somewhat constrained. Various methods have been proposed to tackle the challenge of unsupervised domain adaptation in semantic segmentation. MIC[8] proposed a Masked Image Consistency module for target domain context learning; Context-Aware Domain Adaptation[26] improved context transfer via cross-attention. Yet, domain-specific information integration and uncertainty reduction are still unexplored.

To address these issues, we introduce a novel method based on SAM from a domain adaptation perspective, designed to augment the feature extraction capability and generalization without relying on prompts. We propose the Cross-branch Feature Augmentation Module (CFA) and the Uncertainty-guided Prediction Regularization Module (UPR). CFA incorporates an additional trainable convolutional neural network (CNN) encoder branch, which complements the frozen vision transformer (ViT) encoder, to capture multi-scale and multi-level features. UPR adjusts the normalization layer to promote the adaptation in the endoscopy field and leverages hints to ensure accurate confidence estimation, so as to improve the OOD performance of SAM.

In summary, our primary contributions are as follows: (1) We build a novel SAM-based model named ASPS, to enhance the feature learning capability and domain generalization for polyp segmentation, demonstrating strong performance without the need for prompts. (2) We introduce the Cross-branch

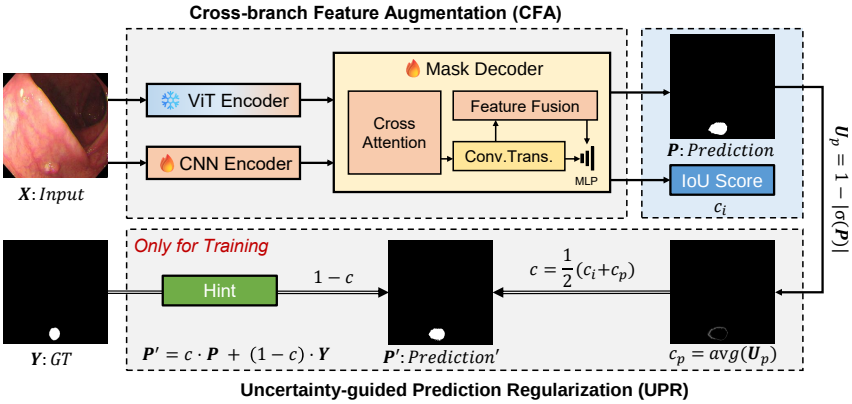

Fig. 1: An overview of our Augmented Segment Anything Model for polyp segmentation. The Cross-branch Feature Augmentation module is encouraged to learn multi-scale features and multi-level representations. The Uncertainty-guided Prediction Regularization module is designed to minimize the uncertainty of the prediction to improve the domain generalization ability of the model.

Feature Augmentation Module (CFA), which introduces an additional CNN encoder branch as a supplement to the ViT encoder. Furthermore, we propose the Uncertainty-guided Prediction Regularization Module (UPR), leveraging hints to reduce uncertainty during training and improve the domain generalization of SAM. (3) Extensive experiments on five common polyp datasets demonstrate the effectiveness and superiority of our method.

2 Method

Overview. Our proposed network is illustrated in Fig. 1. To address the domain degradation issue of SAM, we leverage two modules to enhance its original feature extraction capabilities and domain generalization. The CFA module integrates the CNN encoder feature with global ViT information, leading to generalized feature representation learning. This integration facilitates refined segmentation outputs by aggregating deep information to the superficial layers and incorporating positional information from the shallow levels. Meanwhile, the UPR module is designed to minimize uncertainty and calibrate confidence during training. UPR utilizes a training strategy based on uncertainty, leveraging the ground truth as a guiding 'hint'. The proposed network follows end-to-end training without prompts, jointly optimizing two modules to achieve optimal performance.

2.1 Cross-branch Feature Augmentation Module

While SAM has achieved great success in many image segmentation tasks, it still has some limitations in the polyp segmentation task. One of the main reasons is